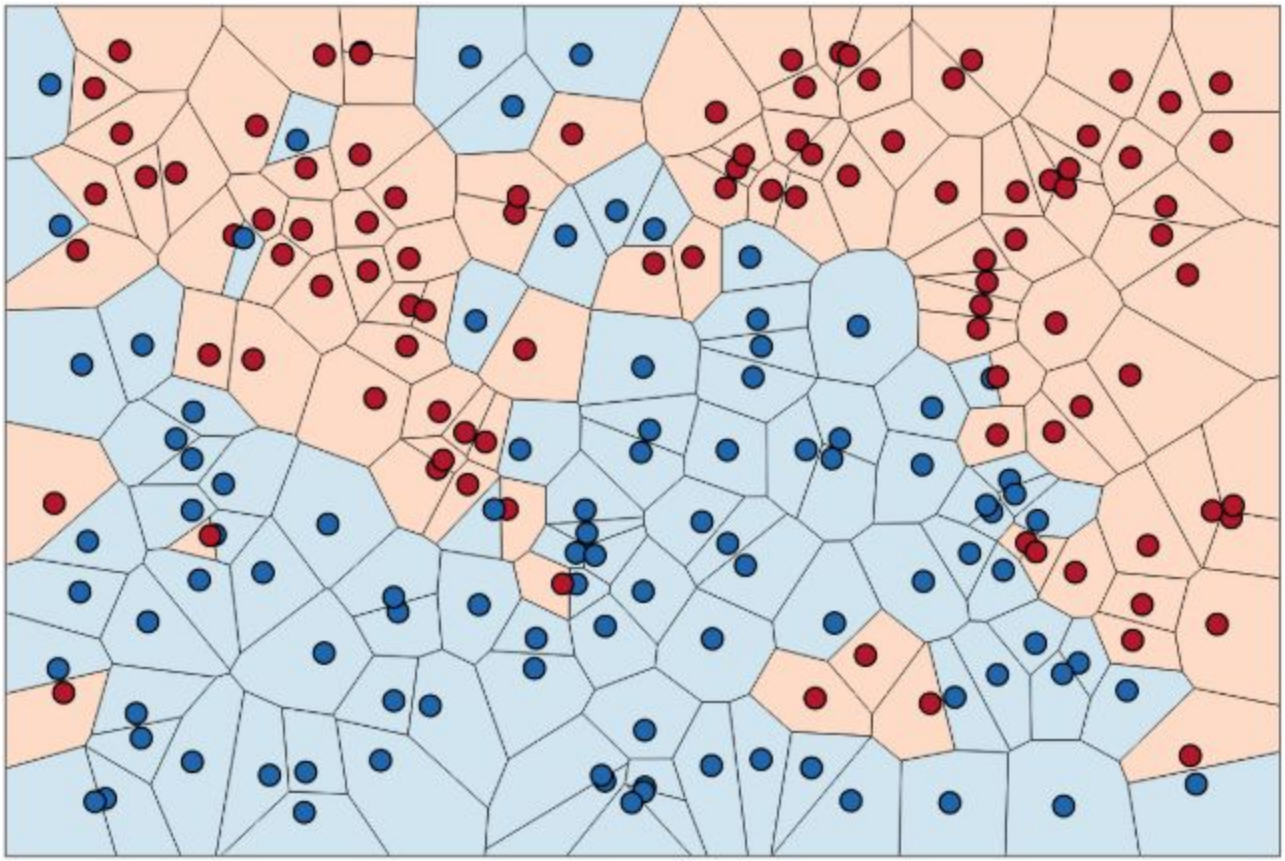
Machine Learning

K Nearest Neighbor

k Nearest Neighbour Algorithm

- Instance-based learning
- Also called Lazy Algorithm
- Find similar instances
- Training Phase
  - Save/store examples in some data structure
- Prediction time
  - Get the test instance
  - Find the training example (x_i, y_i) which is closest to the training example x_t. Then, predict the output y_t.
  - In k nearest neighbour algorithm, take k training example $\{(x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k)\}$ which is closest to the training example x_t.
  - For classification problem: check majority class from $\{y_1, y_2, \ldots, y_k\}$.
  - For regression problem: predict the average of $\{y_1, y_2, \ldots, y_k\}$ as an estimation of y_t.

1 Nearest Neighbour Algorithm: Voronoi Diagram



Distances

Euclidean $$\sqrt{\sum_{i=1}^{k}(x_i - y_i)^2}$$

Manhattan $$\sum_{i=1}^{k}|x_i - y_i|$$

Minkowski $$\left(\sum_{i=1}^{k}(|x_i - y_i|)^q\right)^{1/q}$$

- In the equations, (x_i, y_i) denote two examples
- When we should go for averaging of large k values for regression problem?
  - Noise in attributes
  - Noise in class labels
  - Classes may partially overlapping
- Euclidean distance is applied while equal weights to all attributes on examples.

Distances and k values

- What does it mean by equal weights to all attributes?
  - Scale of attributes are same and also differences
  - Scale attributes to equal range and equal variants
- Small k: captures fine structure of problem space better; may be necessary for small training set
- Large k: Less sensitive to noise (specifically class noise) better problem estimation for discrete classes; larger training set allows use of larger k

Determine k values

- Thumb rule
  - A common heuristic is:

    $k \approx \sqrt{n}$
  - Where, n= number of samples in the dataset.

    Example: If you have 100 samples, try $k \approx 10$

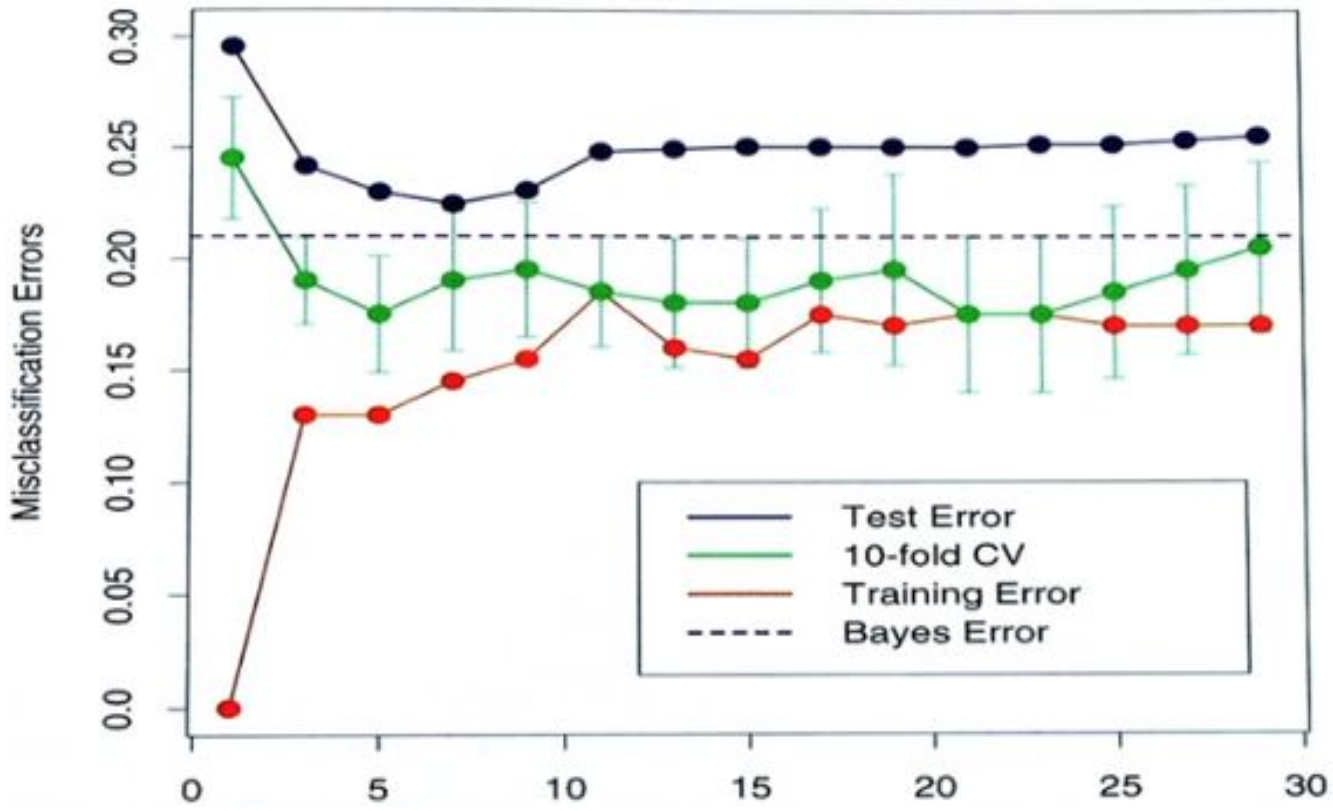
- Cross-validation
  - Split data into training & validation sets (or use k-fold cross-validation).
  - Try different values of k
  - Select the k that gives the lowest error on the validation set.

Determine k values

- Elbow Method
  - Plot validation error against different values of k.
  - Look for the “elbow point” where error stops decreasing significantly.

- Domain knowledge

k Values and Errors



Weighted Euclidean Distance

$$D(c1, c2) = \sqrt{\sum_{i=1}^{N} w_i \cdot \left(attr_i(c1) - attr_i(c2)\right)^2}$$

- large weights => attribute is more important
- small weights => attribute is less important
- zero weights => attribute doesn't matter

Distance Weighted kNN

tradeoff between small and large k can be difficult
– use large k, but more emphasis on nearer neighbors?

$$prediction_{test} = \frac{\sum_{i=1}^{k} w_i * class_i}{\sum_{i=1}^{k} w_i} \left(or \frac{\sum_{i=1}^{k} w_i * value_i}{\sum_{i=1}^{k} w_i} \right)$$

$$w_k = \frac{1}{Dist(c_k, c_{test})}$$

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Locally Weighted Averaging

- Let k = number of training points
- Let weight fall-off rapidly with distance

$$prediction_{test} = \frac{\sum_{i=1}^{k} w_i * class_i}{\sum_{i=1}^{k} w_i} \left(or \frac{\sum_{i=1}^{k} w_i * value_i}{\sum_{i=1}^{k} w_i} \right)$$

$$w_k = \frac{1}{e^{KernelWidth \cdot Dist(c_k, c_{test})}}$$

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Example

We have data from the questionnaires survey (to ask people opinion) and objective testing with two attributes (acid durability and strength) to classify whether a special paper tissue is good or not. Here is four training samples

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Y = Classification
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

Now the factory produces a new paper tissue that pass laboratory test with X1 = 3 and X2 = 7. Without another expensive survey, can we guess what the classification of this new tissue is?

Example

1. Determine parameter K = number of nearest neighbors

Suppose use K = 3

2. Calculate the distance between the query-instance and all the training samples

Coordinate of query instance is (3, 7), instead of calculating the distance we compute square distance which is faster to calculate (without square root)

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	**Square Distance to query instance (3, 7)**
7	7	$(7-3)^2+(7-7)^2=16$
7	4	$(7-3)^2+(4-7)^2=25$
3	4	$(3-3)^2+(4-7)^2=9$
1	4	$(1-3)^2+(4-7)^2=13$

Example

3. *Sort the distance and determine nearest neighbors based on the K-th minimum distance*

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)	**Rank minimum distance**	**Is it included in 3-Nearest neighbors?**
7	7	$(7-3)^2+(7-7)^2=16$	**3**	**Yes**
7	4	$(7-3)^2+(4-7)^2=25$	**4**	**No**
3	4	$(3-3)^2+(4-7)^2=9$	**1**	**Yes**
1	4	$(1-3)^2+(4-7)^2=13$	**2**	**Yes**

4. *Gather the category* Y *of the nearest neighbors.* Notice in the second row last column that the category of nearest neighbor (Y) is not included because the rank of this data is more than 3 (=K).

Example

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)	Rank minimum distance	Is it included in 3-Nearest neighbors?	**Y = Category of nearest Neighbor**
7	7	$(7-3)^2+(7-7)^2=16$	3	Yes	**Bad**
7	4	$(7-3)^2+(4-7)^2=25$	4	No	-
3	4	$(3-3)^2+(4-7)^2=9$	1	Yes	**Good**
1	4	$(1-3)^2+(4-7)^2=13$	2	Yes	**Good**

5. *Use simple majority of the category of nearest neighbors as the prediction value of the query instance*

We have 2 good and 1 bad, since 2>1 then we conclude that a new paper tissue that pass laboratory test with X1 = 3 and X2 = 7 is included in **Good** category.

Advantages and Disadvantages

Advantages

1. The algorithm is simple and easy to implement.
2. There's no need to build a model, tune several parameters, or make additional assumptions.
3. The algorithm is versatile. It can be used for classification, regression, and search (as we will see in the next section).

Disadvantages

1. The algorithm gets significantly slower as the number of examples and/or predictors/independent variables increase.